



AIAP[®] Batch 24 Technical Assessment

Deadline: 1900 hrs, 8th June 2026

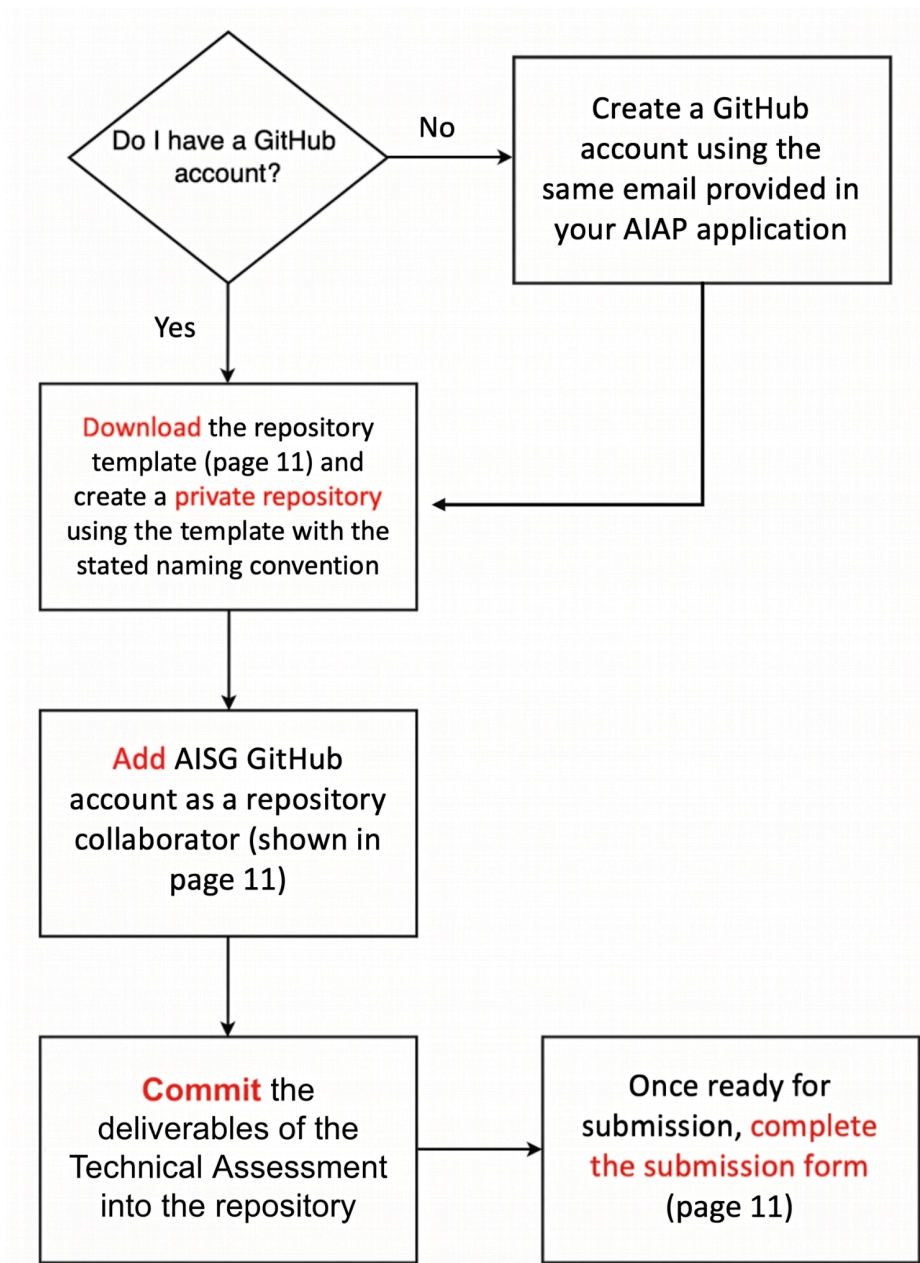
Tasks & Deliverables

This assessment consists of the following parts:

1. Exploratory Data Analysis in Jupyter Notebook
2. Decision Log
3. End-to-end Machine Learning Pipeline in Python Scripts (`.py`)
4. AI Assistant Chat History

Technical Assessment Overview

There are four parts to the Technical Assessment: Decision Log, Exploratory Data Analysis, End-to-end Machine Learning Pipeline, and AI Assistant Chat History. You are to attempt all parts and submit the deliverables by uploading them to your own **private** GitHub repository. The following flowchart outlines the major steps for the Technical Assessment. Details will be provided in the subsequent sections of this document.



Problem Statement

Business Context

MoveEasy is a regional last-mile parcel delivery company operating across Singapore, with four depots (West, North, East, Central) and approximately 200 active drivers — a mix of full-time employees and contracted gig drivers. They handle business-to-consumer deliveries for e-commerce platforms and direct-to-consumer brands. Each delivery is promised within a window — typically 2-hour, same-day, or next-day depending on the customer's plan.

Over the past six months, MoveEasy's average customer rating has dropped from 4.4 to 3.9 stars. Complaints in customer feedback have risen, particularly mentioning late deliveries, rough handling of parcels, and driver behaviour. The CEO is concerned because a major e-commerce client has hinted at switching providers if service quality doesn't improve.

The Head of Operations called your team in. In the conversation, she said:

"Our customer ratings have been dropping and we think it's a delivery problem. We'd like to use AI to fix it. Can your team build us something?"

— Head of Operations, MoveEasy

When pressed for more, she added:

"What I really want is something that helps my dispatch team see, for each upcoming delivery, how likely it is to run into trouble — and ideally, how bad the trouble would be. So they can prioritise their attention. We can't hire more drivers right now, so any solution has to work with the fleet we have."

— Head of Operations, MoveEasy

You have access to MoveEasy's delivery records and customer feedback.

Objectives

Your task is to:

1. Refine these vague requests into a well-defined problem statement that you will work on. Document your thought process in the Decision Log.
2. Perform an Exploratory Data Analysis on the dataset provided.
3. Build an end-to-end machine learning pipeline that addresses the problem statement you defined.
4. Document your use of AI code assistant(s) by submitting a Prompt Chat History alongside your work.

Dataset

You can query the datasets from:

<https://techassessment.blob.core.windows.net/aiap24-assessment-data/delivery.db>

The dataset can be accessed through the `delivery.db`. You may find either of the following packages, `SQLite` or `SQLAlchemy`, useful for accessing this database.

You should place the `delivery.db` file in a `data` folder. Your machine learning pipeline should retrieve the dataset using the relative path `bmarket.db`.

DO NOT upload the `delivery.db` onto your GitHub repository.

Table 1 (Customer Feedback) - List of Attributes

Attribute	Description
feedback_id	Unique feedback identifier
delivery_id	Unique delivery identifier referenced to delivery records. Not all deliveries have feedback.
rating	Customer rating, values ranging for 1 (Bad) to 5 (Good)
comment	Free-text customer comment. May be empty.
feedback_datetime	When the feedback was submitted.

Table 2 (Delivery Records) - List of Attributes

Attribute	Description
delivery_id	Unique delivery identifier
branch	Originating depot (West, North, East, Central)
client_id	Anonymised identifier of client who placed the order
booking_datetime	When the delivery was booked
pickup_datetime	When the parcel was picked up from the depot
promised_delivery_datetime	The delivery time promised to the customer
delivery_datetime	When the delivery was completed (i.e parcel passed to recipient)
distance_km	Distance from depot to recipient
parcel_weight_kg	Weight of the parcel
parcel_value_sgd	Declared insurance value of the parcel, in Singapore dollars
parcel_category	Type of parcel - standard, fragile, oversized, or refrigerated
delivery_priority	Priority tier set by the sender: Standard, Express, Premium, or VIP
num_stops_on_route	Number of stops on the driver's route that day
driver_id	Anonymised driver identifier
driver_experience_months	Driver's tenure with MoveEasy in months
vehicle_type	Type of vehicle used - bike, van, or truck
payment_method	prepaid or cod (cash on delivery)
recipient_postal_code	Anonymised recipient postal area

Submission Requirements — Four Deliverables

Task 1: Exploratory Data Analysis (EDA)

Using the dataset specified in the **Dataset** section, conduct an EDA and create an interactive notebook (.ipynb file) in **Python** that can be used as a presentation to explain the findings of your analysis. It should contain appropriate visualisations and explanations to assist readers in understanding how these elaborations are arrived at and their implications.

Deliverable

- Jupyter Notebook in **Python**: a `.ipynb` file named `eda.ipynb`. (do adhere to the naming requirement)

Evaluation

In the submitted notebook, you are required to

- Outline the steps taken in the EDA process
- Explain the purpose of each step
- Explain the conclusions drawn from each step
- Explain the interpretation of the various statistics generated and how they impact your analysis
- Generate clear, meaningful, and understandable visualizations that support your findings
- Organise the notebook so that it is clear and easy to understand

Please note that your submission will be heavily penalised for any of the following conditions:

- `.ipynb` missing in the submitted repository
- `.ipynb` cannot be opened on Jupyter Notebook
- Explanations missing or unclear in the submitted Jupyter Notebook

Task 2: Decision Log

Complete and submit the `decision\_log.md` file found in the template repository folder (download link below). The template contains five fixed questions:

- Q1 — Clarifying questions
- Q2 — Defining the Problem Statement
- Q3 — Key decisions during the solution development
- Q4 — Use of the AI assistant
- Q5 — Next steps

Deliverable

- `decision\_log.md`

Evaluation

This decision log is the primary instrument by which we understand your thinking. The questions below cover the reasoning behind your work — from how you scoped the problem at the start to the decisions you made during the work itself. Do answer all five questions in your own words. You may use AI assistance freely on the technical deliverables (the EDA and the ML pipeline), but this Decision Log itself should be written by you — it is the record of your own thinking that we cross-check against your chat history.

Task 3: End-to-end Machine Learning Pipeline

Design and create a machine learning pipeline (MLP) in Python scripts (`.py` files) that will ingest and process the entailed dataset, subsequently, feeding it into the machine learning algorithm(s) of your choice.

Do not develop your MLP in an interactive notebook.

The pipeline should be easily configurable to enable easy experimentation of different algorithms and parameters as well as ways of processing data. You can consider the usage of a config file, environment variables, or command line parameters.

Within the pipeline, data must be fetched/imported using SQLite, or any similar packages.

Deliverables

1. A folder named ``src`` containing Python modules/classes in ``py`` format.
2. An executable bash script ``run.sh`` at the base folder of your submission to run the aforementioned modules/classes/scripts. DO NOT install your dependencies in the ``run.sh``; this will be taken care of automatically when we assess the assignment if you have created your ``requirements.txt`` correctly.
3. A ``requirements.txt`` file in the base folder of your submission.
4. A ``README.md`` file that sufficiently explains the pipeline design and its usage. You are required to explain the thought process behind your submitted pipeline in the README. The README is expected to contain the following:
 - Full name (as in NRIC) and email address (stated in your application form).
 - Overview of the submitted folder and the folder structure.
 - Instructions for executing the pipeline and modifying any parameters.
 - Description of logical steps/flow of the pipeline. If you find it useful, please feel free to include suitable visualisation aids (eg, flow charts) within the README.
 - Overview of key findings from the EDA conducted in Task 1 and the choices made in the pipeline based on these findings, particularly any feature engineering. Please keep the details of the EDA in the ``ipynb``. The information in the ``README.md`` should be a quick summary of the details from ``ipynb``.
 - Describe how the features in the dataset are processed (summarised in a table).
 - Explanation of your choice of models for each machine learning task.
 - Evaluation of the models developed. Any metrics used in the evaluation should also be explained.
 - Other considerations for deploying the models developed.

Evaluation

The submitted MLP, including the `README.md`, will be used to assess your understanding of machine learning models/algorithms as well as your ability to design and develop a machine learning pipeline. Specifically, you will be assessed on

- Appropriate data preprocessing and feature engineering
- Appropriate use and optimization of algorithms/models
- Appropriate explanation for the choice of algorithms/models
- Appropriate use of evaluation metrics
- Appropriate explanation for the choice of evaluation metrics
- Understanding of the different components in the machine learning pipeline

In your submitted Python scripts (`.py` files), you will be assessed on the quality of your code in terms of reusability, readability, and self-explanatory.

Please note that your submission will be penalised for any of the following conditions:

- Incorrect format for `requirements.txt`
- `run.sh` fails upon execution
- Poorly structured `README.md`
- Disorganised code that fails to make use of functions and/or classes for reusability
- MLP not submitted in Python scripts (`.py` files), including MLP built using Jupyter Notebooks.

Note for Windows users

DO NOT submit a Windows batch (`.bat`) script in replacement of the bash script. Use either 'Windows Subsystem for Linux (WSL)' or 'Git Bash'/'cygwin' for the creation of the bash script.

Task 4: AI Assistant Chat History

You are permitted to use AI code assistant(s) throughout this assessment. We do not penalise the use of AI; we assess the quality of your use of it.

To support that, complete and submit the ``prompt_chat_history.md`` file from the template repository (download link below). Your submission has two parts:

- Share links — one per AI conversation, for every tool that supports a share-link feature (Claude, ChatGPT, Gemini, etc.).
- Raw, unedited transcripts — produce each conversation into the template, labelling messages as [User] or [AI]. Don't edit, summarise, or rewrite anything afterwards. How you capture these transcripts is your choice — we will not provide guidance on the method. What matters is that the output is faithful to your actual chats.

For tools without share-link support (e.g., GitHub Copilot Chat, Cursor, Claude Code in agent mode), mark the share-link as “not available” and rely on the transcript portion.

Deliverables

- ``prompt_chat_history.md``

Evaluation

In your Prompt Chat History, we assess the quality of your AI use — how you direct the assistant and engage critically with its output. The chat is read alongside your Decision Log to check coherence between the two.

Submission Format

Create a [GitHub](#) account using the **same** email provided in your AIAP application form.

Download the repository template from:

<https://techassessment.blob.core.windows.net/aiap24-assessment-data/aiap24-NAME-NRIC.zip>

The downloaded repository template contains a hidden folder: ``.github``. The ``.github`` folder contains scripts to execute your end-to-end machine learning pipeline using GitHub Actions. Specifically, it will first install the required dependencies using your `requirements.txt` and subsequently, execute your bash script (`run.sh`). You can manually trigger the pipeline under Actions in your repository.

Using the downloaded template, create a **private** repository using the following naming convention:

aiap24-<full name (as in NRIC) separated by dashes>-<last 4 characters of NRIC>

For example, `aiap24-john-lim-der-hui-321A`

Add the following account as a collaborator in your private repository:

- Username: **AISG-AIAP**
- Email: **aiap-internal@aisingapore.org**

Your repository is to have the following structure:

Organise your submission as follows:

```
<your_name>_aiap_submission/  
├── README.md           # pipeline overview, run instructions, EDA summary  
├── eda.ipynb           # Deliverable 2: EDA notebook  
├── src/                # Deliverable 3: ML pipeline code  
├── ...  
├── run.sh              # Deliverable 3: pipeline runner  
├── requirements.txt     # Deliverable 3: dependencies  
├── decision_log.md      # Deliverable 1: completed template  
└── prompt_chat_history.md # Deliverable 4: share links + transcripts
```

Do not include the dataset files in your submission.

We encourage you to adhere to Git best practices. Once your repository is ready for submission, complete the following form at

<https://forms.gle/ugEWsem2wovpWSBS9>

NOTE: You may continue making changes to your repository up to the deadline, even after submitting the Google Form. After the deadline, do not modify your repository — any post-deadline changes will incur a penalty.